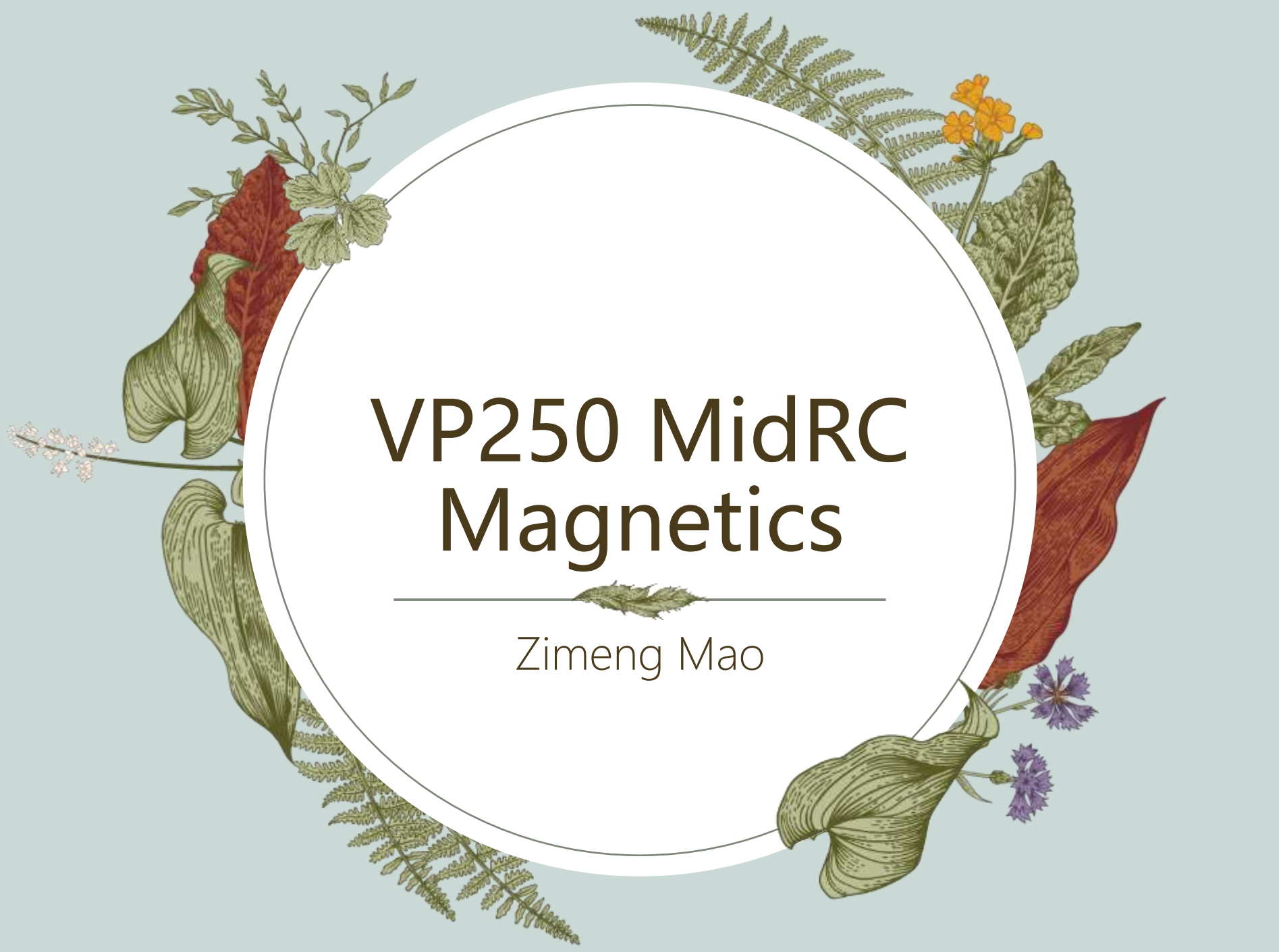


A circular wreath of various botanical illustrations surrounds the central text. The plants include green ferns, a red maple-like leaf, a large green heart-shaped leaf, a cluster of small pink flowers, a large green leaf with prominent veins, a red leaf, and several purple flowers.

VP250 MidRC Magnetics

Zimeng Mao



Magnetic field

Magnetic field due to a point charge with constant velocity

$$\vec{B} = \frac{\mu_0}{4\pi} \frac{q\vec{v} \times \hat{r}}{r^2}$$

Magnetic constant μ_0 **Charge** q **Velocity** $\vec{v}$ **Unit vector from point charge toward where field is measured** $\hat{r}$ **Distance from point charge to where field is measured** r^2





Magnetic flux

Magnetic flux through a surface

Magnitude of magnetic field $\vec{B}$

Component of $\vec{B}$ perpendicular to surface

$$\Phi_B = \int B \cos \phi \, dA = \int B_{\perp} \, dA = \int \vec{B} \cdot d\vec{A}$$

Angle between $\vec{B}$ and normal to surface

Element of surface area

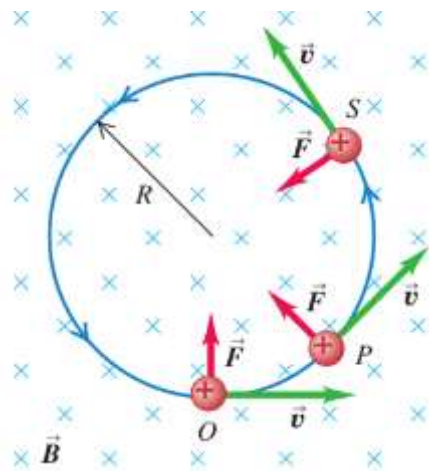
Vector element of surface area

The total magnetic flux through *any* **closed surface** ...

$$\oint \vec{B} \cdot d\vec{A} = 0 \quad \dots \text{equals zero.}$$




Magnetic force



Magnetic force on a moving charged particle

$$\vec{F} = q\vec{v} \times \vec{B}$$

Particle's charge

Magnetic field

Particle's velocity

Radius of a circular orbit in a magnetic field

$$R = \frac{mv}{|q|B}$$

Particle's mass

Particle's speed

Magnetic-field magnitude

Particle's charge



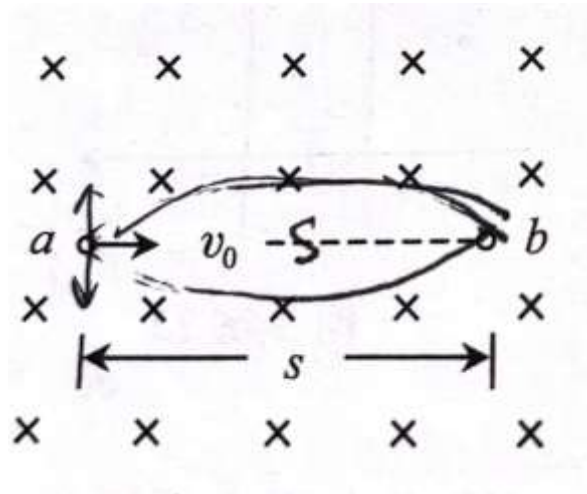


VP27.1.2 A charged particle moving in the presence of a 0.750 T magnetic field experiences a magnetic force of magnitude $2.50 \times 10^{-10}\text{ N}$. The particle is moving at $1.95 \times 10^5\text{ m/s}$ at an angle of 36.0° to the direction of the magnetic field. Find the magnitude of the charge on the particle.





In a uniform and horizontal magnetic field B , two points a and b are on a plane perpendicular to the magnetic field. The distance between a and b is s and the line between ab is on the horizontal plane. A positive electron with mass m and charge q is emitted from a towards b . To make sure that it pass b , what is the value of v_0 ? ←





Magnetic field(current)

Magnetic field due to an infinitesimal current element

$$d\vec{B} = \frac{\mu_0}{4\pi} \frac{I d\vec{l} \times \hat{r}}{r^2}$$

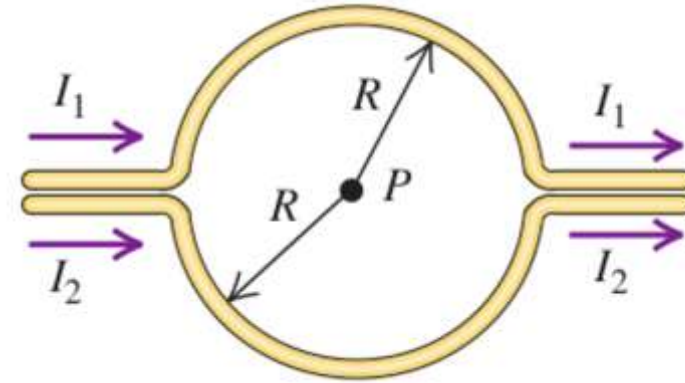
Annotations:

- Magnetic constant (μ_0)
- Current (I)
- Vector length of element (points in current direction) ($d\vec{l}$)
- Unit vector from element toward where field is measured ($\hat{r}$)
- Distance from element to where field is measured (r)





28.33 ●● Calculate the magnitude of the magnetic field at point P of **Fig. E28.33** in terms of R , I_1 , and I_2 . What does your expression give when $I_1 = I_2$?





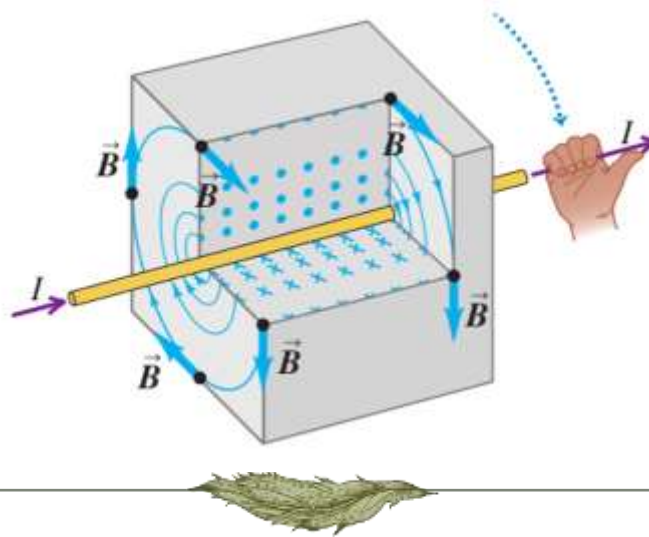
Magnetic field near a long,
straight, current-carrying
conductor

Magnetic constant

$$B = \frac{\mu_0 I}{2\pi r}$$

Current

Distance from conductor



Magnetic field on axis of a circular current-carrying loop

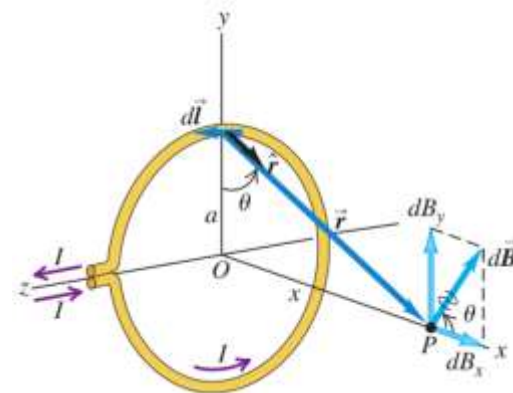
$$B_x = \frac{\mu_0 I a^2}{2(x^2 + a^2)^{3/2}}$$

Magnetic constant

Current

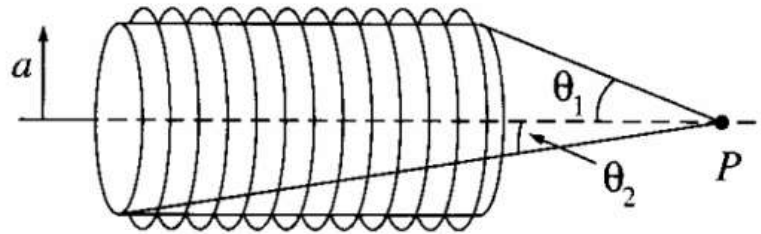
Radius of loop

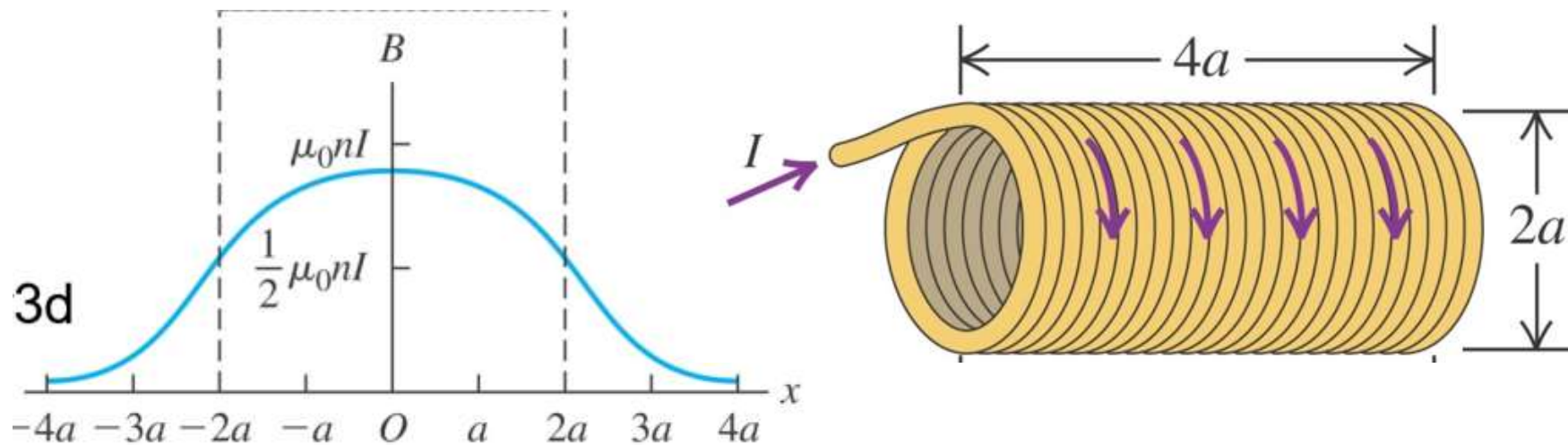
Distance along axis from center of loop to field point





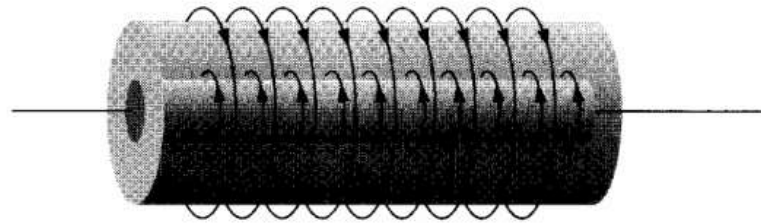
Problem 5.11 Find the magnetic field at point P on the axis of a tightly wound solenoid (helical coil) consisting of n turns per unit length wrapped around a cylindrical tube of radius a and carrying current I (Fig. 5.25). Express your answer in terms of θ_1 and θ_2 (it's easiest that way). Consider the turns to be essentially circular, and use the result of Ex. 5.6. What is the field on the axis of an *infinite* solenoid (infinite in both directions)?







Problem 5.15 Two long coaxial solenoids each carry current I , but in opposite directions, as shown in Fig. 5.42. The inner solenoid (radius a) has n_1 turns per unit length, and the outer one (radius b) has n_2 . Find $\mathbf{B}$ in each of the three regions: (i) inside the inner solenoid, (ii) between them, and (iii) outside both.





Magnetic force

Magnetic force on an infinitesimal wire segment

Current

Magnetic field

$$d\vec{F} = I d\vec{l} \times \vec{B}$$

Vector length of segment (points in current direction)





VP27.7.1 A straight wire 0.150 m in length carries a current of 3.50 A in the $+x$ -direction. The wire is in a uniform 0.0136 T magnetic field in the xy -plane that points in a direction 20.0° from the $-x$ -axis and 70.0° from the $+y$ -axis. Find (a) the magnitude and (b) the direction of the magnetic force on the wire.





Magnetic torque

Magnitude of magnetic torque on a current loop $\tau = IBA \sin \phi$

Current Magnetic-field magnitude
Area of loop Angle between normal to loop plane and field direction

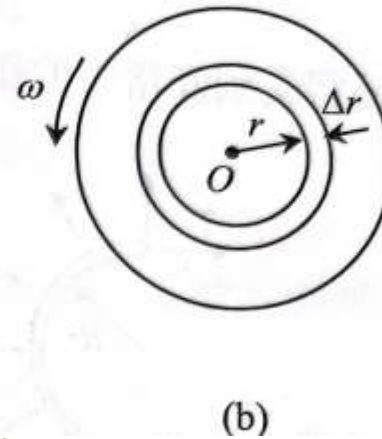
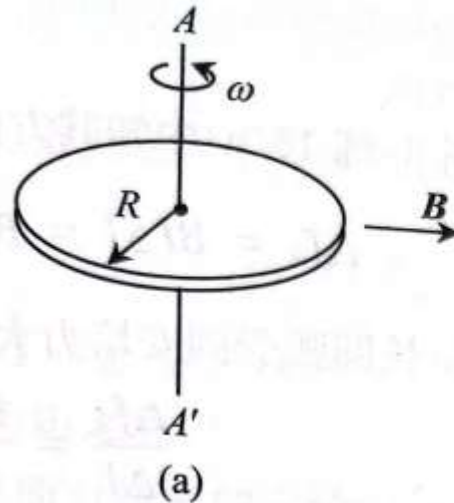
Vector magnetic torque on a current loop $\vec{\tau} = \vec{\mu} \times \vec{B}$

Magnetic dipole moment Magnetic field





2. A plastic disc with radius R , whose surface is uniformly charged σ , is spinning around its axis. The direction of the uniform magnetic field is vertical to its axis. What is the torque of the field on the disc?





Potential energy of magnetic dipole

Potential energy
for a magnetic dipole
in a magnetic field

$$U = -\vec{\mu} \cdot \vec{B} = -\mu B \cos \phi$$

Magnetic dipole moment

Magnetic field

Angle between $\vec{\mu}$ and $\vec{B}$





Ampere's law

Line integral around a closed path

Ampere's law:

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{encl}}$$

Scalar product of magnetic field and vector segment of path

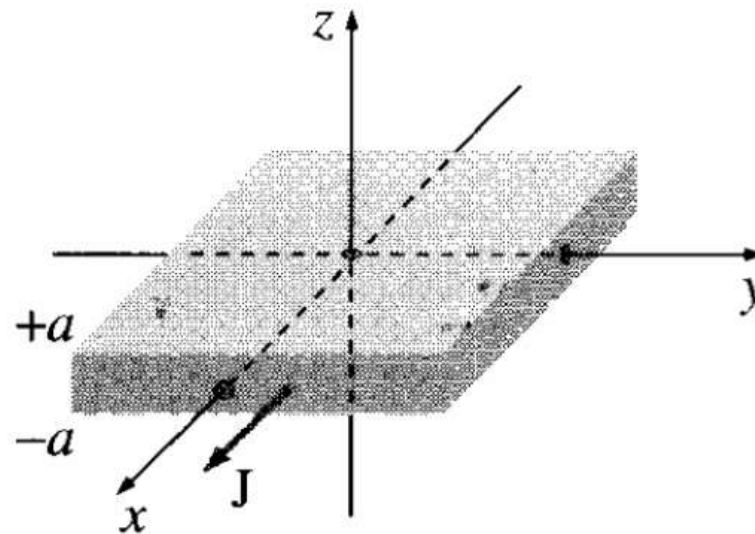
Magnetic constant

Net current enclosed by path





Problem 5.14 A thick slab extending from $z = -a$ to $z = +a$ carries a uniform volume current $\mathbf{J} = J \hat{\mathbf{x}}$ (Fig. 5.41). Find the magnetic field, as a function of z , both inside and outside the slab.



Hall effect

Hall effect:

Concentration of
mobile charge carriers

Charge per carrier

nq

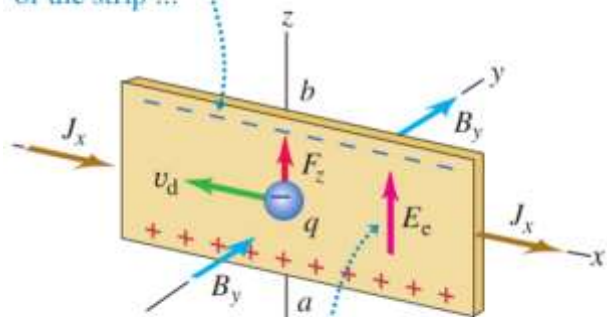
$$nq = \frac{-J_x B_y}{E_z}$$

Current density

Magnetic field

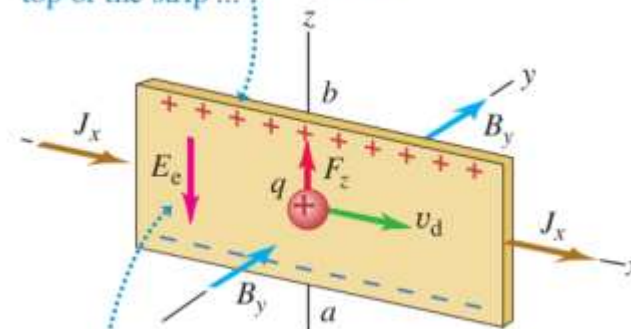
Electrostatic field
in conductor

The charge carriers are pushed toward the top of the strip ...



... so point a is at a higher potential than point b .

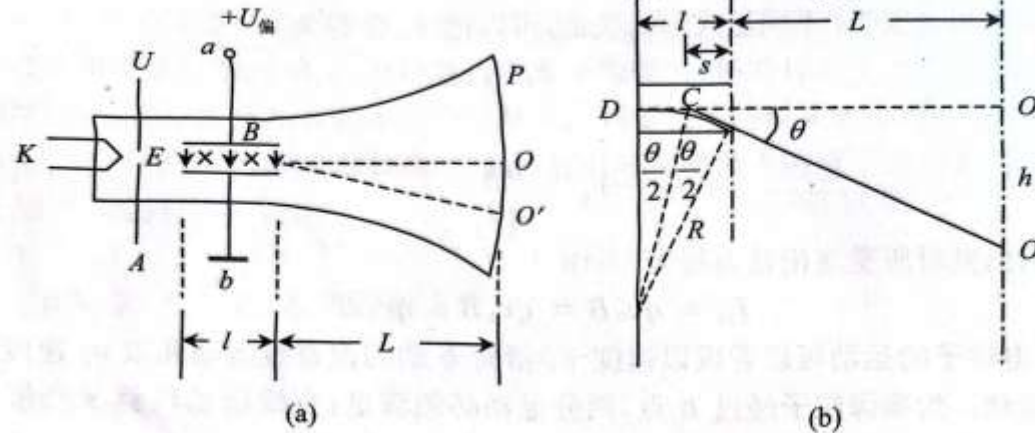
The charge carriers are again pushed toward the top of the strip ...



... so the polarity of the potential difference is opposite to that for negative charge carriers.



An electron is accelerated by a uniform electric field and then enters the magnetic field. The direction of electric and magnetic field is vertical to each other. When first entering, the electron is at equilibrium. Then, remove the electric field and the electron does circular motion in the magnetic field. Other parameters are shown in the picture. Express the ratio of charge and mass of the electron using E B h l and L .↵





Reference

Yijia Yu

- VP250 recitation class, 2023

Professor Wan

- VP250 lecture slides, 2024

Sizhe Sun

- ECE230 recitation class, 2024